Bayesian inference in diffusion tensor imaging

Hugo Eklöf
Master's Programme in Data Science
Uppsala University, Sweden
hugo.eklof.1717@student.uu.se

Metehan Karataş
Department of Information Technology; Division of Computer Systems
Uppsala University, Sweden
metehan.karatas@it.uu.se

Axel Olsson
Master's Programme in Data Science
Uppsala University, Sweden
olsson.axel@hotmail.com

Abstract

Diffusion MRI measures how water moves in tissue, and the diffusion tensor model summarizes this movement in each voxel by a baseline signal S_0 and a 3×3 diffusion tensor $\mathbf{D}$. We study Bayesian inference for this model in a single voxel from the Stanford HARDI dataset, and compare three methods of approximating the posterior. These methods are the Metropolis–Hastings algorithm (MH), variational inference (VI), and the Laplace approximation. We compare the resulting posteriors of S_0, mean diffusivity, fractional anisotropy and the principal diffusion direction. All three methods agree on the direction and on the posterior means. The Laplace approximation reproduces the MH posterior almost exactly, at a few hundred times lower cost. Compared with MH, VI underestimates the uncertainty in S_0 and FA, with 95% credible intervals about half and two thirds as wide, and overestimates it for mean diffusivity, with an interval about 23% wider. Finally, we discuss how each method could be improved and whether it could handle a full scan with around 10^5 voxels. Only the Laplace approximation can do it in a reasonable time.

Contribution statement

Hugo Eklöf: Implemented the frozen prior and frozen likelihood functions. Implemented the code for the Laplace approximation. Wrote the introduction, appendix and the sections about shared methodology, preprocessing and Laplace approximation. Contributed to some sections related to the Laplace approximation in the comparison and discussion section.

Metehan Karataş: Implemented the variational inference method. Implemented the scripts that compare the three methods and produce the comparison figures, Table 1 and the timings. Wrote the abstract, the variational inference section and the comparison section, parts of discussion section and the conclusion.

Axel Olsson: Implemented the code and wrote the section related to the Metropolis-Hasting method. Contributed with section 5.1 in appendix and additional minor work relating to Metropolis-Hasting method in the comparison and discussion section.

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1 Introduction

The diffusion of water molecules in white matter is, contrary to diffusion in free water, not uniform in all directions. This is mainly due to cell membranes and myelin sheaths restricting motion and causing water to more easily diffuse along rather than across axons. By applying magnetic field gradients in different directions and observing how the signal changes, the underlying structure of the white matter can be inferred.

Considering a single voxel of white matter, the diffusion can be described using a symmetric, positive-definite *diffusion tensor* $\mathbf{D}$ given as

$$\mathbf{D} = \begin{pmatrix} D_{aa} & D_{ab} & D_{ac} \\ D_{ab} & D_{bb} & D_{bc} \\ D_{ac} & D_{bc} & D_{cc} \end{pmatrix}, \tag{1}$$

where each entry D_{ij} determines the diffusion along and across different spatial direction. Given a position $\mathbf{r}$, this satisfies

$$S(\mathbf{r}, \mathbf{x}) = S_0(\mathbf{r}) \exp(-\mathbf{x}^\top \mathbf{D}(\mathbf{r})\mathbf{x}), \tag{2}$$

where $\mathbf{x}$ represents the strength and diffusion gradient direction for one measurement and $S_0 \geq 0$ is the baseline signal with $\mathbf{x} = \mathbf{0}$. The recorded measurements y are representative of (2) with added noise of known variance σ^2:

$$p(y \mid \mathbf{r}, \mathbf{x}) = \mathcal{N}(y \mid S(\mathbf{r}, \mathbf{x}), \sigma^2). \tag{3}$$

Given data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ from the Stanford HARDI dataset [3], the goal of the project is to estimate S_0 and $\mathbf{D}$ for a single voxel.

2 Shared methodology and preprocessing

The problem can be described using Bayes' theorem

$$p(\mathbf{z} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mathbf{z})p(\mathbf{z})}{p(\mathcal{D})}, \tag{4}$$

where $\mathbf{z} = \{S_0, \mathbf{D}\}$ are our latent variables. The goal is to find the posterior $p(\mathbf{z} \mid \mathcal{D})$, which will be done by using *three* different Bayesian inference methods: The Metropolis-Hastings algorithm, variational inference and Laplace approximation. Common for all three methods is that they make use of the same prior $p(\mathbf{z})$ and likelihood $p(\mathcal{D} \mid \mathbf{z})$, which with the eigendecomposition $\mathbf{D} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ (A.1) are given as

$$p(\mathbf{z}) = p(S_0)\, p(\mathbf{\Lambda})\, p(\mathbf{V}), \tag{5}$$

$$p(\mathcal{D} \mid \mathbf{z}) = \Pi_{i=1}^N \mathcal{N}(y_i \mid S(\mathbf{r}, \mathbf{x}_i), 29^2) \tag{6}$$

$$S_0 \sim \text{Gamma}(2, 500), \tag{7}$$

$$\lambda_1, \lambda_2, \lambda_3 \sim \text{Gamma}(4, 2.5 \cdot 10^{-4}), \tag{8}$$

$$\mathbf{V} \sim \text{Uniform}\,(\text{SO}(3)), \tag{9}$$

where the hyperparameters of the distributions are given and $p(\mathbf{\Lambda}) = p(\lambda_1)\, p(\lambda_2)\, p(\lambda_3)$ due to eigenvalue independence. To avoid numerical underflow, the log-probabilities of the prior and likelihood are used:

$$\log p(\mathbf{z}) = \log p(S_0) + \sum_{i=1}^{3} \log p(\lambda_i), \tag{10}$$

$$\log p(\mathcal{D} \mid \mathbf{z}) = \sum_{i=1}^{N} \log \mathcal{N}(y_i \mid S(\mathbf{r}, \mathbf{x}_i), 29^2). \tag{11}$$

Note that the constant $\log p(\mathbf{V})$ is excluded from (10) since constants are not needed for our purposes. The implementations of both log-probabilities were validated using given data before the Bayesian inference methods were applied. As point estimate, and as the starting point for all three methods, we use the weighted least-squares fit of the diffusion tensor model from `dipy.reconst.dti`.

3 The Metropolis-Hastings algorithm

To approximate the posterior distribution using the Metropolis-Hastings algorithm, the aim is to obtain samples from a Markov chain whose invariant distribution is the posterior, as the number of iterations $m \to \infty$. The method requires the unnormalized posterior distribution

$$\pi(\mathbf{z}) \propto p(\mathcal{D} \mid \mathbf{z})p(\mathbf{z}), \tag{12}$$

where $p(\mathcal{D} \mid \mathbf{z})$ is the likelihood and $p(\mathbf{z})$ is the prior (4). Additionally, a proposal distribution $q(\mathbf{z}' \mid \mathbf{z})$ is predefined to generate a new candidate sample $\mathbf{z}'$ given the current state

$$q(\mathbf{z}' \mid \mathbf{z}) = \text{Gamma}(\gamma^{-2}, \gamma^2 S_0) \cdot \mathcal{W}(\mathbf{D}, \nu) \tag{13}$$

To ensure that the Markov chain has the desired invariant distribution, the transition probabilities are constructed to satisfy equality between forward and reverse sampling.

$$\pi(\mathbf{z})q(\mathbf{z}' \mid \mathbf{z})\alpha(\mathbf{z}, \mathbf{z}'),) = \pi(\mathbf{z}')q(\mathbf{z} \mid \mathbf{z}')\alpha(\mathbf{z}', \mathbf{z}),. \tag{14}$$

where $\alpha(\mathbf{z}, \mathbf{z}')$ is Metropolis-Hastings acceptance probability of a new sample $\mathbf{z}'$ given the current $\mathbf{z}$.

$$\alpha(\mathbf{z}, \mathbf{z}') = \min\left(1, \frac{\pi(\mathbf{z}')q(\mathbf{z} \mid \mathbf{z}')}{\pi(\mathbf{z})q(\mathbf{z}' \mid \mathbf{z})}\right). \tag{15}$$

The acceptance probability is compared with a random sample $u \sim \mathcal{U}(0, 1)$. If $u < \alpha(\mathbf{z}, \mathbf{z}')$, the proposed state is accepted; otherwise, the chain remains at its current state.

The hyperparameters ν and γ in the proposal distribution (13) were tuned to $\nu = 2000$ and $\gamma = 0.01$, resulting in an acceptance of 0.48, balancing convergence speed and exploration of the posterior domain. The chain was initialized using a point estimate from the given model. The implementation was done in log-space for numerical stability, with $50\,000$ samples generated, of which the first $m = 500$ were discarded as burn-in, as the trace plots (A.5) indicated that the sampled variables had reached their stationary distributions by that point. The final distributions are visualized in 1.

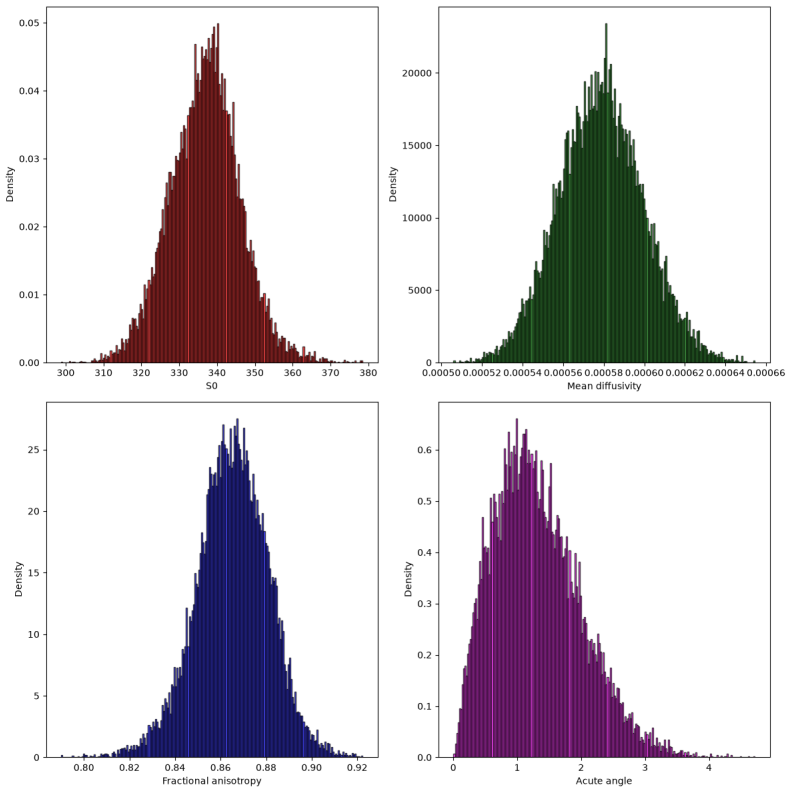


Figure 1: Histograms representing distibution of S_0, mean diffusivity, fractional anisotropy and acute angle.

4 Variational inference

Since the posterior cannot be sampled from directly, VI approximates it by the member of a tractable family $\mathcal{Q}$ that is closest in KL divergence, $q^*(\mathbf{z}) = \arg\min_{q\in\mathcal{Q}} \mathrm{KL}\big(q(\mathbf{z}) \,\|\, p(\mathbf{z} \mid \mathcal{D})\big)$, found by maximizing the ELBO. We use the variational posterior

$$q(z) = q(S_0)\, q(D), \quad q(S_0) = \Gamma(\alpha, \beta), \quad q(D) = \mathcal{W}(D, \nu), \tag{16}$$

with the reparameterization $\alpha = e^{\theta_1}$ and $\beta = e^{\theta_2}$ to ensure positivity. The Cholesky factorization $\mathbf{D} = \mathbf{L}(\theta_{3:8})\mathbf{L}^\top(\theta_{3:8})$ (A.2) was used to ensure positive definiteness of the diffusion tensor and $\nu = e^{\theta_9} + 2$ to ensure that the Wishart distribution had at least two degrees of freedom.

As in the provided `variational_posterior` class, a sample is $\mathbf{D} = \mathbf{W}/\nu$ with $\mathbf{W} \sim \mathcal{W}(\mathbf{L}\mathbf{L}^\top, \nu)$, so $\mathbf{L}\mathbf{L}^\top$ is the mean of $q(\mathbf{D})$ and ν only sets its spread. The provided analytical score $\nabla_{\boldsymbol{\theta}} \log q_{\boldsymbol{\theta}}(\mathbf{z})$ did not match finite differences of $\log q_{\boldsymbol{\theta}}$ in θ_9. It holds $\mathbf{W}$ fixed, but for a fixed sample $\mathbf{D}$, $\mathbf{W} = \nu\mathbf{D}$ changes with ν. Adding the missing term made the two agree. The prior (5) is written in terms of the eigenvalues and eigenvectors of $\mathbf{D}$, but q is written in terms of $\mathbf{D}$ itself. The ELBO compares the two, so they must be densities over the same variable. We therefore convert the prior to a density over $\mathbf{D}$ (a change of variables) and for each sampled $\mathbf{D}$ compute its eigenvalues and subtract $\sum_{j<k} \log|\lambda_j - \lambda_k|$ from the log-prior.

We maximize the ELBO with Adam for 10^4 iterations, using the REINFORCE leave-one-out estimator (A.3) with $K = 20$ Monte Carlo samples. The off-diagonal entries $\theta_4, \theta_6, \theta_7$ enter $\mathbf{L}$ directly and are of order 10^{-3}, so they get a step size of $5 \cdot 10^{-4}$, while the log-scale parameters use 0.05; the step size decays exponentially to 5% of its initial value. These settings, including K, were chosen by trial and error, keeping the combination that gave the highest and most stable ELBO; for space reasons, only the final run is shown. Since the variance of the gradient estimate falls roughly as $1/K$ while the cost grows linearly in K, smaller K gave a noticeably noisier ELBO, and larger K increased the run time without noticeable improvement.

We start at the point estimate: $\mathbf{L}\mathbf{L}^\top = \hat{\mathbf{D}}$, $\nu = 100$, and $q(S_0)$ centered on $\hat{S}_0$ with variance $\sigma^2 / \sum_i A_i^2$, $A_i = \exp(-\mathbf{x}_i^\top \hat{\mathbf{D}} \mathbf{x}_i)$, the inverse curvature of the log-likelihood in S_0 for fixed $\mathbf{D}$.

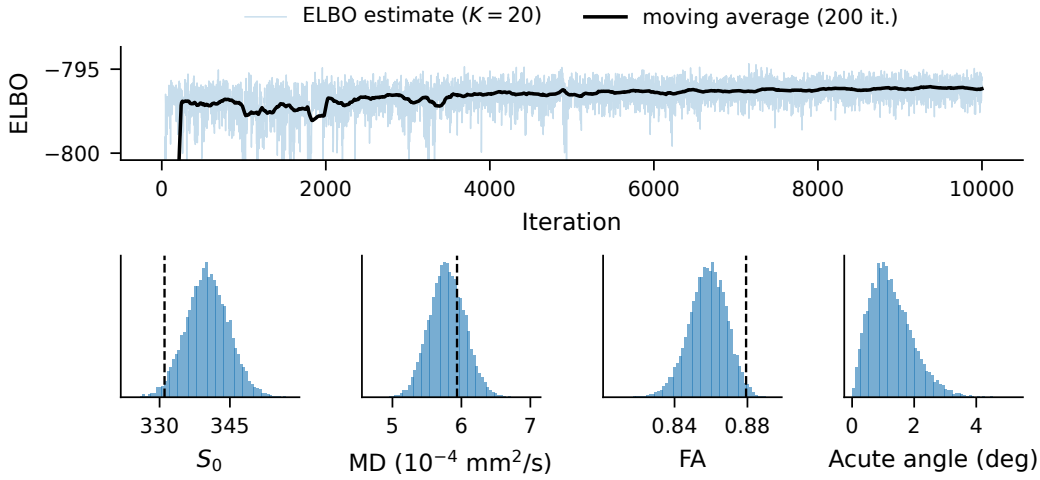


Figure 2: Variational inference. Top: ELBO estimate per iteration and its 200-iteration moving average. Bottom: histograms of the baseline signal S_0, mean diffusivity, fractional anisotropy and acute angle φ from 10^4 samples of the fitted variational posterior. Dashed lines mark the point estimate.

The ELBO (Figure 2, top) climbs from about -808 to -797.5 in the first 10^3 iterations and levels off at around -796, with little change in the moving average over the last $5 \cdot 10^3$ iterations. The resulting posterior summaries are given in Table 1.

5 Laplace approximation

The Laplace approximation approximates the posterior $p(\mathbf{z} \mid \mathcal{D})$ by fitting a Gaussian distribution centered at its mode $\mathbf{z}_0$ [1]. The method considers an unnormalized distribution $\tilde{p}(\mathbf{z} \mid \mathcal{D})$ satisfying

$$p(\mathbf{z} \mid \mathcal{D}) = \frac{1}{C} \cdot \tilde{p}(\mathbf{z} \mid \mathcal{D}), \tag{17}$$

where C is a normalization constant. The mode $\mathbf{z}_0$ then satisfies

$$\nabla_{\mathbf{z}} \log \tilde{p}(\mathbf{z} \mid \mathcal{D}) \mid_{\mathbf{z}=\mathbf{z}_0} = 0. \tag{18}$$

By taking the 2nd degree Taylor expansion of $\log \tilde{p}(\mathbf{z} \mid \mathcal{D})$ centered on $\mathbf{z}_0$, we get

$$\log \tilde{p}(\mathbf{z} \mid \mathcal{D}) \approx \log \tilde{p}(\mathbf{z} \mid \mathcal{D})|_{\mathbf{z}=\mathbf{z}_0} + \frac{1}{2}(\mathbf{z}-\mathbf{z}_0)^\top \nabla^2 \log \tilde{p}(\mathbf{z} \mid \mathcal{D})|_{\mathbf{z}=\mathbf{z}_0}(\mathbf{z}-\mathbf{z}_0). \tag{19}$$

Letting $-\Sigma^{-1} := \nabla^2 \log \tilde{p}(\mathbf{z} \mid \mathcal{D})|_{\mathbf{z}=\mathbf{z}_0}$, (19) can be written as

$$\tilde{p}(\mathbf{z} \mid \mathcal{D}) \approx \tilde{p}(\mathbf{z} \mid \mathcal{D})|_{\mathbf{z}=\mathbf{z}_0} \cdot \exp\left\{-\frac{1}{2}(\mathbf{z}-\mathbf{z}_0)^\top \Sigma^{-1}(\mathbf{z}-\mathbf{z}_0)\right\}. \tag{20}$$

Comparing this result with the standard result for a normalized multivariate Gaussian (A.4), we conclude that

$$p(\mathbf{z} \mid \mathcal{D}) \approx \mathcal{N}(\mathbf{z} \mid \mathbf{z}_0, \Sigma). \tag{21}$$

This method was applied using a transformed space similar to the previous method to ensure positivity, with $S_0 = e^{\theta_1}$ and the Cholesky factorization $\mathbf{D} = \mathbf{L}(\theta_{2:7})\mathbf{L}^\top(\theta_{2:7})$. The mode $\hat{\boldsymbol{\theta}}$ was found by minimizing the *unnormalized* negative log posterior w.r.t. $\boldsymbol{\theta} \in \mathbb{R}^7$ using the BFGS-algorithm, with the given point estimates of S_0 and $\mathbf{D}$ as initial values. The Hessian Σ was subsequently calculated using finite-difference approximation. After this, samples $\boldsymbol{\theta}_n$ could be taken directly from $\mathcal{N}(\mathbf{z} \mid \mathbf{z}_0, \Sigma)$ to produce the histograms in Figure 3.

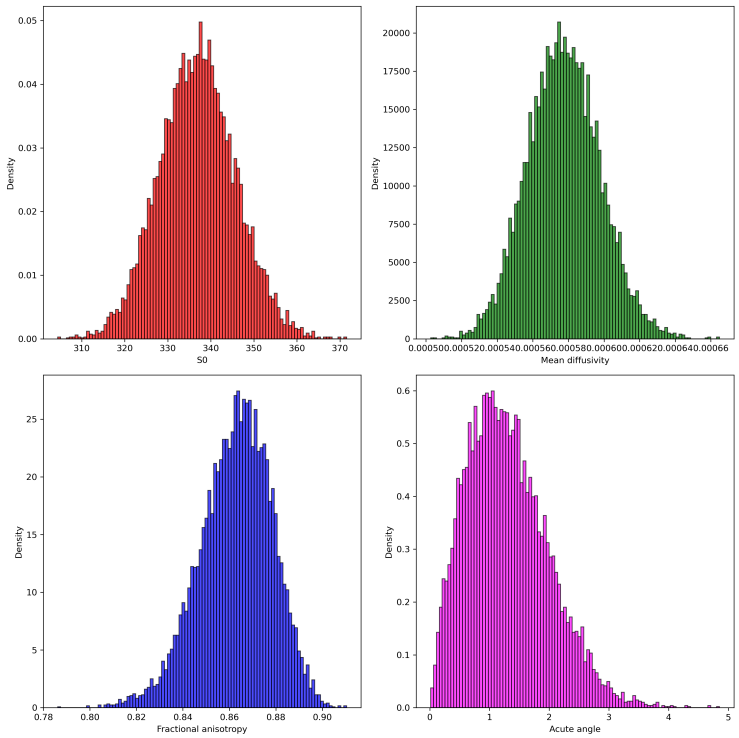


Figure 3: Histograms depicting distributions of S_0, mean diffusivity, fractional anisotropy and acute angle based on samples generated from the Laplace approximation.

6 Comparison and discussion

Table 1: Posterior mean (sd) and central 95% credible interval. MH: $5.0 \cdot 10^4$ samples after $5 \cdot 10^2$ burn-in; VI and Laplace: 10^4 samples. MD in 10^{-4} mm^2/s, angle in degrees. Point estimate: $S_0 = 331.1$, MD $= 5.94$, FA $= 0.879$.

	Metropolis–Hastings (MH)		Variational inference (VI)		Laplace	
	mean (sd)	95% CI	mean (sd)	95% CI	mean (sd)	95% CI
S_0	337.3 (9.04)	[320.1, 355.6]	340.2 (4.5)	[331.5, 349.3]	336.9 (9.0)	[319.7, 354.6]
MD	5.78 (0.21)	[5.37, 6.19]	5.79 (0.26)	[5.30, 6.31]	5.77 (0.21)	[5.37, 6.19]
FA	0.864 (0.016)	[0.829, 0.893]	0.858 (0.010)	[0.837, 0.877]	0.863 (0.016)	[0.830, 0.891]
Angle	1.31 (0.68)	[0.23, 2.86]	1.27 (0.69)	[0.23, 2.85]	1.30 (0.69)	[0.22, 2.88]

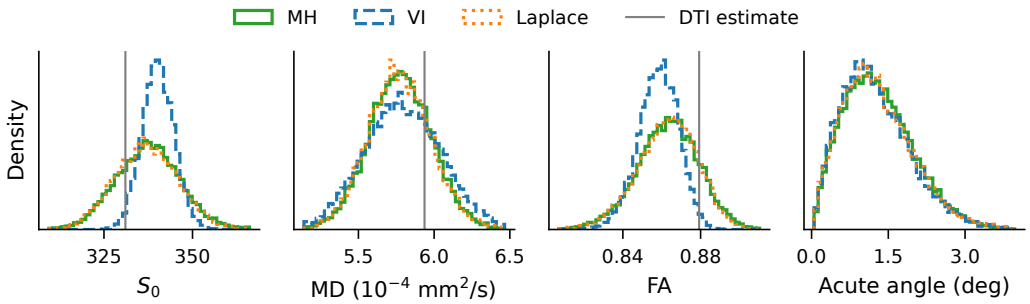


Figure 4: Marginal posteriors of S_0, mean diffusivity, fractional anisotropy and acute angle for the three methods. The gray line marks the point estimate.

Table 1 and Figure 4 show the posteriors from the three methods side by side. All three use the same model and data, and all start from the same point estimate. The table gives the mean and standard deviation of these values, and a 95% credible interval, which is the range between the 2.5% and 97.5% percentiles of the samples. MH targets the exact posterior, but only once the chain has converged, which is assumed to be the case after the discarded burn-in period. Even though the effective sample size is significantly lower than the samples from the generated chain, the assumption is still made that enough samples were realized under convergence to conclude that MH is sufficiently close to the true posterior for it to be used as a reference for the two approximations.

All three methods agree on the orientation of the tensor: the acute-angle distributions are practically identical, with a median of 1.2°. Laplace is also very close to MH for everything else. The means differ by at most 0.1 sd, the widths of the 95% intervals by at most 4%, and the correlations between S_0, MD and FA by at most 0.02. The only visible difference is a slightly shorter upper tail for FA. The posterior is clearly close to Gaussian. The point estimate lies about 0.7–1 sd from the MH and Laplace means, along the direction where S_0 and FA trade off against each other (Figure 5). It falls inside the MH and Laplace 95% intervals, but just outside the VI intervals for S_0 and FA.

For VI, its 95% intervals for S_0 and FA are only about half and two thirds as wide as those of MH, while its interval for MD is about 23% too wide. Both problems come from the structure of VI, not from the optimization being executed poorly.

In the true posterior, S_0 and $\mathbf{D}$ are clearly correlated (correlation $+0.65$ with MD and -0.64 with FA under MH). The factorized q cannot represent this, and VI then ends up with a $q(S_0)$ that is as narrow as if $\mathbf{D}$ were known.

The Wishart factor has a single parameter ν that sets the relative spread of all three eigenvalues at once. The true posterior is much less certain about the two small eigenvalues than about the largest one, because diffusion across the fiber gives only a weak signal. VI settles for a compromise, although it is clear setting all three equal makes for a very bad fit. Both of these results can be seen in the right part of Figure 5.

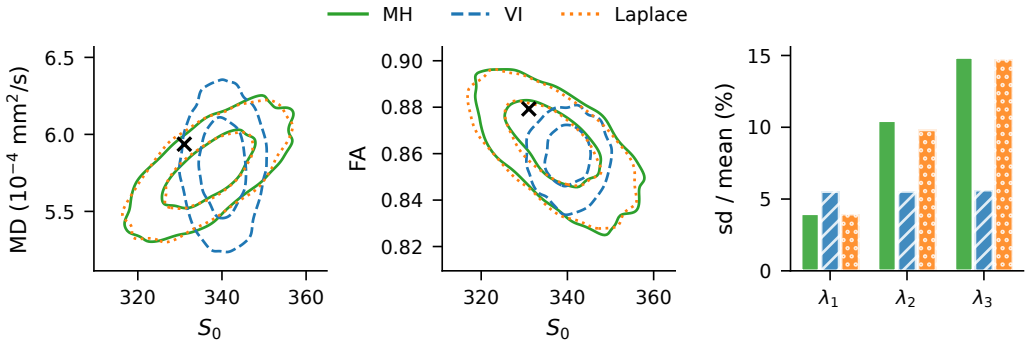


Figure 5: Left, middle: 50% and 90% highest-density (KDE) contours of the joint posteriors; the cross marks the weighted least-squares DTI point estimate (Section 2). Right: relative spread (sd/mean) of the ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$.

VI gives an explicit density that is cheap to sample from (10^4 samples in 0.1 s), and the ELBO can be used to compare models. However, it can never be better than the chosen distribution family, and the noisy REINFORCE gradient needed a lot of fine tuning.

The Laplace approximation was by far the fastest and, for this voxel, as accurate as MH. One of the models strengths is that it is expected to better approximate the posterior as the number of observed data points is increased due to the Central Limit Theorem [1]. Its weakness is that it assumes a single, roughly Gaussian mode and gives no warning when that fails, for example in noisy or nearly isotropic voxels where the orientation is poorly determined. A possible improvement would be to use a weighted sum of multiple Laplace approximations if the true distribution would have been multimodal.

A key strength of the Metropolis–Hastings algorithm is its ability to converge to the target distribution. However, a limitation is that the generated samples are correlated, making it difficult to assess the uncertainty of the estimates and determine the number of samples required to achieve sufficient accuracy. Furthermore, the method requires considerable domain knowledge for an effective implementation, including appropriate initialization, the selection of a suitable proposal distribution, and the choice of a burn-in period.

With our unoptimized Python code on a laptop CPU (Mac M3), one voxel takes about 0.06 s with Laplace, 20 s with MH ($5 \cdot 10^4$ iterations) and 50 s with VI (10^4 iterations). For a full scan with $\sim 10^5$ voxels, that is about 1.7 hours for Laplace, 23 days for MH and 60 days for VI on a single core. All methods can be parallelized over voxels, but only Laplace is practical as it is.

7 Conclusion

We compared MH, VI and the Laplace approximation on the same model and voxel. All three agree on the posterior means and on the principal diffusion direction, so for point estimates the choice of method matters little; they differ in how they represent uncertainty. The Laplace approximation reproduces the MH posterior, with 95% intervals within 4% in width, at about $1/300$ of the cost, which indicates that the posterior for this strongly anisotropic voxel is close to Gaussian. VI misrepresents the uncertainty: its intervals for S_0 and FA are about half and two thirds as wide as those of MH, while its interval for MD is about 23% wider. These errors follow from the variational family, namely the factorization between S_0 and $\mathbf{D}$ and a single ν for all eigenvalues, rather than from the optimization, so more iterations or samples would not remove them. For a full scan, only the Laplace approximation is fast enough as implemented. Since it gives no warning when the posterior is not Gaussian, it should be checked against MH on a few voxels of other types, such as noisy or nearly isotropic ones.

Use of generative AI

We used generative AI as a support tool in the following ways. To help find errors in our implementation in the code and to write the scripts that produce the comparison figures and the numbers. To revise and simplify our drafts and to help with LaTeX formatting. We have read, checked and edited everything the AI produced, and we take full accountability/ownership for the content of this report.

References

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[2] Roger A Hom and Charles R Johnson. Matrix analysis. *Cambridge University Express*, 455, 1985.

[3] Ariel Rokem, Jason D. Yeatman, Franco Pestilli, Kendrick N. Kay, Aviv Mezer, Stefan van der Walt, and Brian A. Wandell. Evaluating the accuracy of diffusion mri models in white matter. *PLOS ONE*, 10(4):1–26, 04 2015.

[4] Jiaxin Shi, Yuhao Zhou, Jessica Hwang, Michalis Titsias, and Lester Mackey. Gradient estimation with discrete stein operators. *Advances in neural information processing systems*, 35:25829–25841, 2022.

A Appendix

A.1 Eigendecomposition

Let $\mathbf{D}$ be a symmetric, positive-definite 3×3 diffusion tensor with eigenvalues $\lambda_1, \lambda_2, \lambda_3 \geq 0$. Then the eigendecomposition of $\mathbf{D}$ is given as

$$\mathbf{D} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top, \tag{22}$$

where

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \tag{23}$$

and $\mathbf{V} \in \mathrm{SO}(3)$ is a rotation matrix.

A.2 Cholesky factorization

The Cholesky factorization of a symmetric, positive-definite matrix $\mathbf{D}$ is given as

$$\mathbf{D} = \mathbf{L}\mathbf{L}^\top, \tag{24}$$

where L is a real lower triangular matrix with positive diagonal elements [2]. For this report, the triangular matrix $\mathbf{L}$ takes the form

$$\mathbf{L}(\theta_{1:6}) = \begin{pmatrix} e^{\theta_1} & & \\ \theta_2 & e^{\theta_3} & \\ \theta_4 & \theta_5 & e^{\theta_6} \end{pmatrix}. \tag{25}$$

A.3 REINFORCE leave-one-out estimator

Let $q_\theta(\mathbf{z})$ be a variational distribution and define

$$f(\mathbf{z}) = \log p(\mathcal{D}, \mathbf{z}) - \log q_\theta(\mathbf{z}). \tag{26}$$

Then the gradient of the expected objective can be written as

$$\nabla_\theta \mathbb{E}_{q_\theta}\left[f(\mathbf{z})\right] = \mathbb{E}_{q_\theta}\left[f(\mathbf{z}) \nabla_\theta \log q_\theta(\mathbf{z})\right] \tag{27}$$

$$\approx \frac{1}{K} \sum_{k=1}^{K} \left(f(\mathbf{z}^{(k)}) - \frac{1}{K-1} \sum_{j \neq k} f(\mathbf{z}^{(j)}) \right) \nabla_\theta \log q_\theta(\mathbf{z}^{(k)}), \tag{28}$$

where $K \geq 2$ is the number of Monte Carlo samples [4].

A.4 Multivariate Gaussian distribution

For a vector $\mathbf{x}$ of dimension D, the multivariate Gaussian distribution takes the form

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\mathbf{\Sigma}|^{1/2}} \exp\left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^{\mathrm{T}} \mathbf{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

where $\boldsymbol{\mu}$ is a D-dimensional mean vector, $\mathbf{\Sigma}$ is a $D \times D$ covariance matrix, and $|\mathbf{\Sigma}|$ denotes the determinant of $\mathbf{\Sigma}$ [1].

A.5 Trace plots for the Metropolis-Hasting algorithm

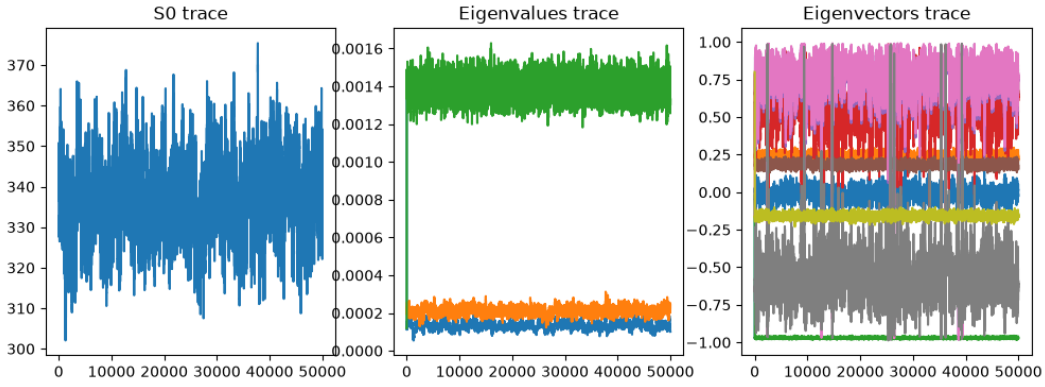


Figure 6: Trace plots for the Metropolis-Hastings algorithm

The figure illustrates the trace plots of the Metropolis-Hastings Markov chain, S_0, the three eigenvalues $\lambda_1, \lambda_2, \lambda_3$, and the three entries of each of the three eigenvectors. The y-axis shows the parameter value, and the x-axis shows the sample index.